Probability Theory and Random Processes (MA225)

Lecture Slides

Lecture 26

Classification of States

Def: For a MC $\{X_n : n \geq 0\}$, the first passage time to state i is the random variable T_i defined by

$$T_i = \inf \{n \geq 1 : X_n = i\},$$

where $\inf \emptyset = \infty$.

Def: A state i is called recurrent if $P_i(T_i < \infty) = 1$.
Def: A state i is called transient if $P_i(T_i < \infty) < 1$.

Remark: By strong Markov property, i is recurrent if and only if

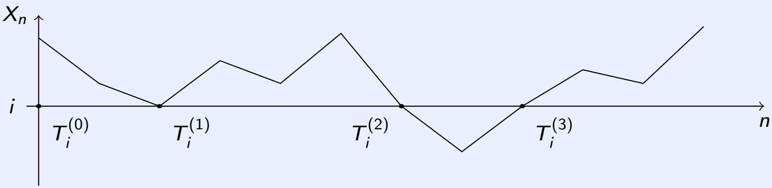
$$P_i(X_n = i \text{ for infinitely many } n) = 1.$$

kth Passage Time and Excursions

Def: For a state i, define $T_i^{(0)} = 0$ and for $k \geq 0$

$$T_i^{(k+1)} = \begin{cases} \inf\left\{n : n > T_i^{(k)}, X_n = i\right\} & \text{if } T_i^{(k)} < \infty \\ \infty & \text{otherwise.} \end{cases}$$

$T_i^{(k)}$ is called the kth passage time to state i.



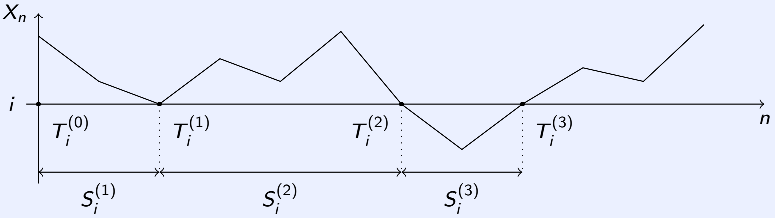
Remark: If i is a recurrent state, then for all $k \geq 0$,

$$P_i\left(T_i^{(k)} < \infty\right) = 1.$$

kth Passage Time and Excursions

Def: The length of the kth excursion to i is given by

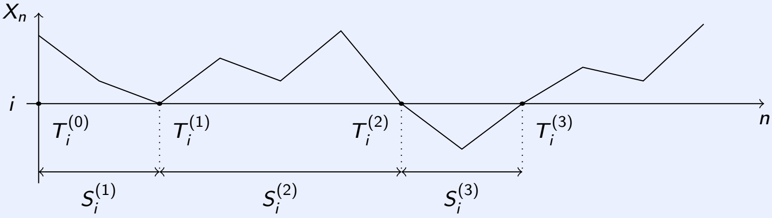
$$S_i^{(k)} = \begin{cases} T_i^{(k)} - T_i^{(k-1)} & \text{if } T_i^{(k-1)} < \infty \\ 0 & \text{otherwise.} \end{cases}$$



Distribution of $S_i^{(k)}$

Lemma: For $k = 2, 3, \ldots$, conditional on $T_i^{(k-1)} < \infty$, $S_i^{(k)}$ is independent of $\left\{X_m : m \leq T_i^{(k-1)}\right\}$ and

$$P\left(S_i^{(k)} = n \middle| T_i^{(k-1)} < \infty\right) = P_i\left(T_i = n\right).$$



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Proof:

- We will use strong Markov property at the stopping time $T = T_i^{(k-1)}$.
- It is easy to see that $X_T = i$ on $T < \infty$.
- Thus, conditional on $T < \infty$, $\{X_{T+n}\}_{n \geq 0}$ is a MC with same TPM and initial distribution δ_i and it is independent of $X_0, X_1, \ldots, X_T$.
- From definition,

$$S_i^{(k)} = \inf\left\{n \geq 1 : X_{T+n} = i\right\}.$$

- So, $S_i^{(k)}$ is the first passage time of $\{X_{T+n}\}_{n \geq 0}$ to state i.

Number of Visits

Def: For a state i, we define the number of visits to i by

$$V_i = \sum_{n=0}^{\infty} \delta_i(X_n).$$

Distribution of V_i

Lemma: For $k = 0, 1, 2, \ldots$, we have $P_i(V_i > k) = f_i^k$, where $f_i = P_i(T_i < \infty)$.

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Lemma: For $k = 0, 1, 2, \ldots$, we have $P_i(V_i > k) = f_i^k$, where $f_i = P_i(T_i < \infty)$.

Proof:

- We will prove the lemma by induction.
- Observe that $\{V_i > k\} = \left\{T_i^{(k)} < \infty\right\}$.
- The statement is true for $k = 0$.
- Suppose that it is true for k.
- Then

$$\begin{aligned} P_i(V_i > k+1) &= P_i\left(T_i^{(k+1)} < \infty\right) = P_i\left(T_i^{(k)} < \infty,\, S_i^{(k+1)} < \infty\right) \\ &= P_i\left(S_i^{(k+1)} < \infty | T_i^{(k)} < \infty\right) P_i\left(T_i^{(k)} < \infty\right) \\ &= f_i f_i^k = f_i^{k+1}, \end{aligned}$$

  where the second last equality follows by the previous lemma and induction hypothesis.
- This completes the proof.

Transient (Revisited)

Corollary: A state i is transient if and only if

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Proof: Since i is transient, $f_i < 1$. Thus,

$$\begin{aligned} P_i\left(X_n = i \text{ for infinitely many } n\right) &= P_i\left(V_i = \infty\right) \\ &= \lim_{k \to \infty} P_i\left(V_i > k\right) \\ &= 0. \end{aligned}$$

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Remark: Recall that i is recurrent if and only if

$$P_i\left(X_n = i \text{ for infinitely many } n\right) = 1.$$

Equivalent Conditions for Recurrent and Transient

Theorem:

1. A state i is recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.
2. A state i is transient if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$.

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Proof:

- We only prove part 1. Part 2 follows from part 1.
- Suppose that i is recurrent.
- Then $f_i = 1$.
- Hence,

$$P_i\left(V_i = \infty\right) = \lim_{k \to \infty} P_i\left(V_i > k\right) = 1.$$

- So,

$$\infty = E_i(V_i) = E_i\left(\sum_{n=0}^{\infty} \delta_i(X_n)\right) = \sum_{n=0}^{\infty} E_i\left(\delta_i(X_n)\right) = \sum_{n=0}^{\infty} P_i\left(X_n = i\right) = \sum_{n=0}^{\infty} p_{ii}^{(n)}.$$

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2. A state i is transient if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$.

Proof:

- Now, Suppose that $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.
- If possible suppose i is transient.
- Then $f_i < 1$.
- So,

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = E_i\left(V_i\right) = \sum_{r=0}^{\infty} P_i\left(V_i > r\right) = \sum_{r=0}^{\infty} f_i^r = \frac{1}{1-f_i} < \infty,$$

  which is a contradiction.
- Hence, i is recurrent.

[Fact: For a non-negative integer valued random variable X, $E(X) = \sum_{n=0}^{\infty} P\left(X > n\right)$.]

Theorem

Theorem: If $\{X_n : n \geq 0\}$ is a MC with finite state space S, then at least one state must be recurrent.

Proof:

- Let $S = \{1, 2, \dots, N\}$.
- Note that $n = \sum_{i=1}^{N} \sum_{j=1}^{n} \delta_i(X_j)$.
- Thus, taking $n \to \infty$ on both sides, we get that there must exist an i_0 such that $\sum_{j=1}^{\infty} \delta_{i_0}(X_j) = \infty$ with positive probability.
- This implies that i_0 must be recurrent.